Designing a Teacher Dashboard for WearableLearning

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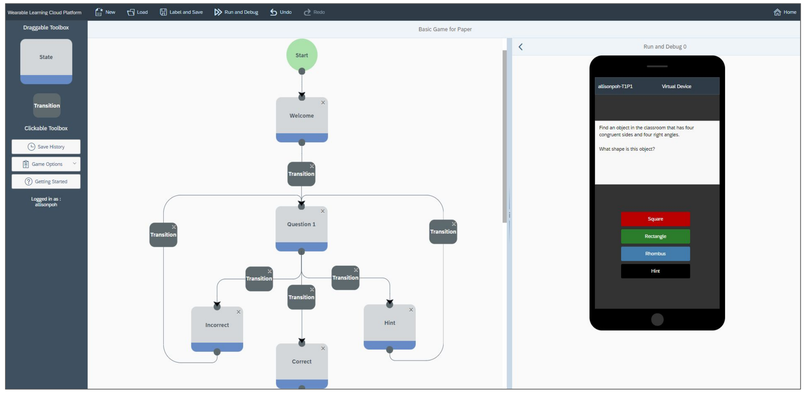
With the goal of enhancing K-12 education through technology, this research focuses on designing a teacher dashboard for WearableLearning, informed by empirical research into the current support learning technologies provide to educators.

What is WearableLearning?

WearableLearning (WL) is a learning technology that promotes math and computational thinking in K-12 education through **game design** and **embodied gameplay**. Games are created using WL's visual programming language, which are then "compiled" to run on Android devices that students typically wear on armbands.



How Does WL Currently Support Teachers?

While students are playing games, teachers have minimal support. Teachers can:

- Start and end a game
- View which students have joined a game

Teachers **cannot**:

- Monitor student progress
- Gain insight into learning
- Intervene immediately if necessary

This highlights the need for WL to provide better support and insightful information to teachers, enabling them to make informed decisions while monitoring their students in real-time. **The goal of this research is to determine how to best build teacher support into WearableLearning.**

Gaining Empirical Insights into Teacher Support in Learning Technologies

To explore how teachers perceive the supportiveness of learning technologies and to determine how teacher support can be better integrated into WL, we adopted a multi-method approach involving 3 parts: (1) a survey, (2) interviews, and (3) a questionnaire.

(1) Survey:

Who? 84 current K-12 teachers

What? 8 questions: 3 demographic and 5 open-ended about learning technologies used in real classrooms and teachers' perceptions of their supportiveness

Data analysis? Word clouding, affinity diagramming, sentiment analysis

Insights? 7 categories of *current* support (instruction, communication, personalized learning, student success, information & insights, community, usability) and 5 categories of *needed* support (availability, consistency, professional development, student monitoring, additional learning opportunities)

(2) Interviews:

Who? 2 current K-12 teachers

What? 2 rounds of interviews: 1st focusing on teachers' perceptions of supportiveness, and 2nd focusing on the supportiveness of a WL dashboard mock-up

Insights? Participants echoed many of the findings from the survey, with emphasis on:

- *Information & insight*: want immediate information that is understandable at a glance
- *Student monitoring*: want to see students' exact screens at any moment and be flagged to students needing intervention

(3) Questionnaire:

Who? 8 prospective K-12 teachers

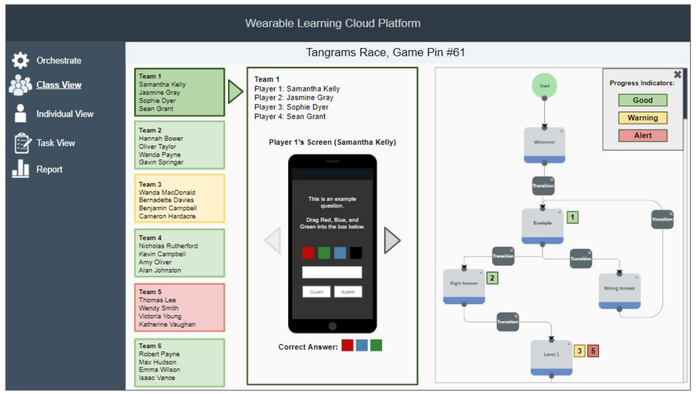
What? 5 open-ended questions regarding WL and how it can be improved to better support teachers

Why? To supplement not having many teacher participants in the interview rounds

Insights? Again, participants echoed the survey findings, with emphasis on:

- *Information & insight*: want stored information for reflection and planning
- *Instruction*: want easy-to-follow instructions

Results: WL Teacher Dashboard Prototype



Next Steps

Evaluate the prototype:

- Cognitive walkthrough
- In-class observation

Then redesign and reevaluate (iterative design process).

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